B.Tech Petroleum Engineering

MS: Mid Semester | IA: Internal Assessment | ES: End Semester | LW: Laboratory Work | LE/Viva: Laboratory Examination/Viva | CE: Continuous Evaluation

Semester I													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24MA101T	Mathematics-I	3	1	0	4	4	**25**	**25**	**50**	--	--	100
24CP101T	Computer Programming-I	1	0	0	1	1	25	25	50	--	--	100
24CP101P	Computer Programming-I Laboratory	0	0	2	2	1	--	--	--	50	50	100
24PH103T	Modern Physics	3	0	0	3	3	25	25	50	--	--	100
24PH103P	Modern Physics Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CV101T	Environmental Science	2	0	0	2	2	25	25	50	--	--	100
24ME101P	Workshop Practice	0	0	2	2	1	--	--	--	50	50	100
24ME102P	Engineering Graphics	0	0	4	4	2	--	--	--	50	50	100
24PE101T	Geology for Petroleum Engineers	3	0	0	3	3	25	25	50	--	--	100
24PE101P	Geology for Petroleum Engineers Practical	0	0	2	2	1	--	--	--	50	50	100
24HS101T	English Communication	2	0	2	2	2	25	25	50	--	--	100
24HS102T	Universal Human Values	1	0	0	1	1	25	25	50	--	--	100
	Total	**15**	**1**	**12**	**28**	**22**						

Semester II													
Course Code	**Course Name**	**Theory**	**Tutorial**	**Practical**	**Hrs**	**Credits**	**Examination Scheme**					
							Theory			**Practical**		**Total**
							MS	**IA**	**ES**	**LW**	**LE/Viva**	
24MA103T	Mathematics-II	3	1	0	4	4	25	25	50	--	--	100
24CP102T	Computer Programming-II	1	0	0	1	1	25	25	50	--	--	100
24CP102P	Computer Programming-II Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH102T	Chemistry	3	0	0	3	3	25	25	50			100
24CH102P	Chemistry Practical	0	0	2	2	1	--	--	--	50	50	100
24PE102T	Sedimentary and Petroleum Geology	3	0	0	3	3	25	25	50	--	--	100
24PE102P	Sedimentary and Petroleum Geology Practical	0	0	2	2	1	--	--	--	50	50	100
24HS103T	Indian Knowledge System	2	0	0	2	2	25	25	50	--	--	100
24HS104T	Organizational Behaviour	1	0	0	1	1	25	25	50	--	--	100
24HS105T	Professional Communication	2	0	0	2	2	25	25	50	--	--	100
24YOG101	Yoga, Health & Hygiene OR NCC/NSS											
24NSS101	National Service Scheme (NSS)	0	0	2	2	1	--	--	--	50	50	100
24NCC101	National Cadet Corps (NCC)											
	Total	**15**	**1**	**8**	**24**	**20**						

24MA103T					Mathematics – II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	-	-	100

COURSE OBJECTIVES

- Understand and apply principles of complex differentiation and integration effectively.
- Solve systems, find eigenvalues, and apply transformations confidently.
- Solve various ODEs, apply methods, and tackle engineering problems.
- Analyze periodic functions, derive series, and apply in diverse applications.

Unit I: Complex Differentiation **Hours: 14**

Limit, Continuity, Differentiability of the function of a complex variable, Analytic function, Cauchy-Riemann equation (in Cartesian coordinates), Harmonic function and its significance, Singularities, Taylor's series, Definition of a Complex line integral, Contour integrals, Cauchy integral theorem, Cauchy Integral formula (CIF), CIF for derivatives, Calculation of residues, Cauchy Residue theorem, Applications of residues to evaluate real definite integrals

Unit II: Matrix Algebra and its Applications **Hours: 14**

Solution of system of algebraic equation - Rank of a matrix, consistency of system of equation - Characteristic equation of a square matrix- Eigen values and Eigenvectors of a real matrix - Properties of Eigen values and Eigen vectors - Cayley-Hamilton theorem (without proof) - finding inverse of a matrix - Diagonalization of a matrix using orthogonal transformation.

Unit III: Ordinary Differential Equations with Applications **Hours: 12**

Differential equations of the first order and higher degree, Higher-order differential equations with constant coefficients, Rules for finding C.F. and P.I., Method of variation of parameters, Cauchy and Legendre's linear equations, Linear differential equations of second order with variable coefficients; Simultaneous linear equations with constant coefficients, applications of higher-order differential equations in solving engineering problems.

Unit IV: Fourier Series **Hours: 12**

Periodic functions, Odd and even functions, Euler's formulae for Fourier series in an interval of length 2 pi, Change of interval, Dirichlet's conditions, Half range Sine and Cosine series, Complex Fourier series, Parseval's identity and its applications.

Max <52Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall fundamental concepts of complex differentiation, matrix algebra, differential equations, and Fourier series.
CO2 - Understand the significance of complex differentiation, integration, Matrix, ODE and Fourier series in respective contexts.
CO3 - Apply the concept of complex function, Matrix, ODE and Fourier series to extract the solutions of engineering problems
CO4 - Analyze the use of complex variable Matrix, ODE and Fourier series in engineering problems.
CO5 - Assess the significance and effectiveness of mathematical concepts and theorems in solving real-world problems, particularly in engineering and scientific applications.
CO6 - Design and construct solutions to complex mathematical problems using a variety of techniques, including transformations, mappings, and advanced methods in differential equations.

TEXT / REFERENCE BOOKS

1. Churchill, R. V., & Brown, J. W. (2003). *Complex variables and applications* (7th ed.). McGraw-Hill.
2. Howie, J. M. (2003). *Complex analysis*. Springer-Verlag.
3. Jain, R. K., & Iyengar, S. R. K. (2007). *Advanced engineering mathematics* (3rd ed.). Alpha Science International.
4. Kreyszig, E. (2015). *Advanced engineering mathematics* (10th ed.). John Wiley & Sons.
5. Strang, G. (2005). *Linear algebra and its applications* (4th ed.). Cengage Learning.
6. Hoffman, K., & Kunze, R. A. (2002). *Linear algebra* (2nd ed.). Prentice Hall of India.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24CP102T					Computer Programming – II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

- To impart the basic concepts of Python Programming language.
- To be familiar with data structures available in Python.
- To understand the concept of function & object-oriented programming.
- To use the file and exception handling for designing applications

Unit I: Introduction and Language Fundamentals **Hours: 04**

Definition and its historical background, Comparison with other programming languages. Python IDEs Tokens: Identifiers, Keyword Data Types: Int, Float, Bool, Complex, Str, List, Tuple, Dictionary and Set, Type conversion, Concept of mutability, immutability and reusability Operators: Arithmetic, Relational, Logical, Identity and Membership, Input and Output methods Control Flow Conditional statements: If, elif and else, nesting, Iterative or loop statement: for, while, for else and while else, Transfer statements: break, continue, pass.

Unit II: Functions **Hours: 03**

Function Declaration and calling, Types of Function arguments, scope of variable, Recursive function Recursive function Types of function: Anonymous or lambda, Map, Filter, Reduce, Function aliasing, Nested function, Decorator function, Decorator chaining Generator function

Unit III: Object Oriented Programming **Hours: 03**

Class, Object, Object reference, Constructor and self-variable, Types of variables, Types of method, Destructor, Composition, Aggregation, Inheritance, Polymorphism, Abstract classes.

Unit IV: File and Exception Handling **Hours: 03**

File handling Types of files, writing and reading with text file, important functions of file handling, working with directories, CSV file handling, pickling and unpickling Exception Handling exceptions handling using try-except blocks. Raising exceptions and custom exception classes.

Max <13Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basic concepts of programming with python.
CO2 - Demonstrate proficiency in using different data types, operators, and control structures in Python programs.
CO3 - Demonstrate the usage of both built-in and user-defined functions.
CO4 - Define and use classes in Python to represent attributes and methods, and demonstrate composition, aggregation, inheritance and polymorphism.
CO5 - Implement exception handling mechanisms to deal with errors.
CO6 - Demonstrate file operations and exception handling.

TEXT / REFERENCE BOOKS

1. Guttag, J. V. (2013). *Introduction to computation and programming using Python*. Prentice Hall.
2. Downey, A. B., Elkner, J., & Meyers, C. (2002). *How to think like a computer scientist: Learning with Python*. Green Tea Press.
3. Sweigart, A. (2015). *Automate the boring stuff with Python: Practical programming for total beginners*. No Starch Press.
4. Brown, M. C. (2018). *Python: The complete reference* (4th ed.). McGraw-Hill Education.
5. Rao, R. N. (2018). *Core Python programming*. Dreamtech Press.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24CP102P					Computer Programming-II Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To impart the basic concepts of Python programming language.
- To be familiar with data structures available in Python.
- To understand the concept of function & object-oriented programming.
- To use file and exception handling for designing applications

LIST OF EXPERIMENTS

1. **Language fundamentals:** Understanding interpretation, program execution flow, tokens, variables and keywords and identifiers, constants and variables
2. **Input Output Operations:** Basic data types, mutability, immutability, Operators, Input /output statements, format specifiers, escape sequences
3. **Decision making, Looping control structures**: If, elif and else, nesting, Iterative or loop statement: for, while, for else and while else, Transfer statements: break,
4. **Advanced datatypes:** String, List, Tuples, Dictionary, Set
5. **Functions:** Basics of functions, types of arguments
6. **Types of functions**: Anonymous function, function aliasing, nested function, decorator function and generator function
7. **Object Oriented programming:** class, object, types of variable and methods, composition, aggregation, inheritance, types of inheritance
8. **File Handling**: writing and reading with text file and csv file, working with directories, pickling and unpickling

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Apply the concepts of variables, decision making and looping with python
CO2 - Apply various data structures available in Python to solve computational problems
CO3 - Implement advanced function concepts such as decorator chaining and generator functions.
CO4 - Design and implement object-oriented solutions to programming problems using composition, aggregation, inheritance and polymorphism
CO5 - Develop Python scripts, for file handling tasks.
CO6 - Create and manage files in python.

TEXT / REFERENCE BOOKS

1. Guttag, J. V. (2013). *Introduction to computation and programming using Python*. Prentice Hall.
2. Downey, A. B., Elkner, J., & Meyers, C. (2002). *How to think like a computer scientist: Learning with Python*. Green Tea Press.
3. Sweigart, A. (2015). *Automate the boring stuff with Python: Practical programming for total beginners*. No Starch Press.
4. Brown, M. C. (2018). *Python: The complete reference* (4th ed.). McGraw-Hill Education.
5. Rao, R. N. (2018). *Core Python programming*. Dreamtech Press.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24CH102T					Chemistry					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- To provide a broad foundation in chemistry that stresses scientific reasoning and analytical problem solving with a molecular perspective
- To provide students with the skills required to succeed in graduate school, the chemical industry or professional school.
- To expose the students to a breadth of experimental techniques using modern instrumentation

Unit I: Physical Chemistry **Hours: 10**

Surface Chemistry: Interparticle forces, adsorption isotherms, determination of the surface area of fine powders using BET theory, surface films; Colligative Properties and their Experimental Determination: Boiling Point Elevation, Freezing Point depression, Osmotic Pressure.

Unit II: Organic Chemistry **Hours: 10**

Alkanes, Alkenes, Alkynes; Markovnikov's rule; Peroxide effect; Bayer's test; Monohydric alcohols; Saytzeff rule; Methods of distinguishing the three classes of alcohols; Aldol condensation; Clemmensen reduction; Wolf-Kishner reduction; Haloform reaction; Cannizzaro reaction; Reformatsky reaction; Wittig reaction; Saturated monocarbozylic acids; Hell-Volhard-Zelinsky reaction; Classification of carbohydrates; Monosaccharides; Mutarotation; Epimerization; Aromatics; Ruff degradation Aromaticity; Huckel rule; Electrophilic substitution reactions; Directive effects of substituents; Aromatic amino compounds ; Carbylamines reaction

Unit III: Petroleum Chemistry **Hours: 10**

Solid fuels: Wood, charcoal, Peat, Coal, Analysis of coal, ASTM classification, Ash, Coal and its environment, fluidized bed combustion and coke. Liquid Fuels: Diesel Engine, Diesel Fuel, Ignition temperature, flash point, fire point and smoke point, The spark Ignition internal combustion engine, gasoline fuel. Gaseous fuel: Natural gas and its uses, Natural gas as a fuel, other carbon based fuel gases, and explosion limits. Hydrogen, methods for the preparation of H_2, Transportation and storage of H_2

Unit IV: Organic Geochemistry and Isotope Geochemistry **Hours: 09**

Organic Geochemistry; A brief Biological background, Organic Compound and their nomenclature, Biologically important organic compounds; Carbohydrate, Lipids, Proteins, Introduction to isotope geology; Discovery or radioactivity and isotopes; Isotope in earth sciences; Nuclide types and their abundances; Decay mechanism of radioactive atoms; basic Principles of radiometric dating; Stable isotope Systematic: elementary knowledge about fractionation of stable isotopes; Stable isotopes and petroleum; mass spectrometry.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basic concept of surface chemistry including analysis of BET surface area and colligative properties.
CO2 - Illustrate the fundamental concepts relevant to organic reaction mechanism
CO3 - Analyse the synthesis and properties of Hydrocarbons (alkane, alkenes and alkynes), monohydric alcohol, monosaccharides, amines, amino acids and aromatic compounds.
CO4 - Demonstrate the mechanism of important name reactions.
CO5 - Explain the basic and fundamental concept of solid, liquid and gaseous fuels
CO6 - Evaluate the principles of organic and isotope geochemistry including important biologically organic compounds.

TEXT / REFERENCE BOOKS

1. Finar, I. L. (2002). *Organic chemistry* (Vol. 1, 6th ed.). Pearson Education.
2. Atkins, P. (2006). *Physical chemistry* (8th ed.). Oxford University Press.
3. Roussak, O. V., & Gesser, H. D. (2013). *Applied chemistry: A practical textbook for engineers and technologists* (2nd ed.). Springer.
4. Faure, G., & Mensing, T. M. (2005). *Isotopes: Principles and applications* (3rd ed.). Wiley.
5. Killops, S. D., & Killops, V. J. (2005). *Introduction to organic geochemistry* (2nd ed.). Blackwell Publishing.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24CH102P					Chemistry Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To provide a broad foundation in chemistry that stresses scientific reasoning and analytical problem solving with a molecular perspective.
- To provide students with the skills required to succeed in graduate school, the chemical industry or professional school.
- To expose the students to a breadth of experimental techniques using modern instrumentation.

LIST OF EXPERIMENTS

1. Determination of the concentration (strength) of a given sodium hydroxide solution by titrating it against a standard solution of hydrochloric acid
2. Determination of total temporary and permanent hardness of a given water samples
3. Determination of the Biological Oxygen Demand (BOD) of a given water sample
4. Determination of the Chemical Oxygen Demand (COD) of a given water sample
5. Determination of viscosity of a given liquid by Ostwald viscometer
6. Adsorption Studies – Langmuir and Freundlich Adsorption Isotherms
7. Determination of Saponification value of an oil
8. Determination the moisture & volatile contents in a given coal sample by proximate analysis
9. Identification of alcoholic group in organic sample
10. Identification of keto group in organic sample
11. Identification of Phenolic OH determination in organic sample
12. Identification of Amines in organic sample
13. Identification of carboxylic group (-COOH) in organic sample
14. Determination of salinity and chlorinity of sea water

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Interpret different types of instruments for qualitative and quantitative analysis of chemicals.
CO2 - Identify the organic functional groups in a given sample.
CO3 - Determine the physio-chemical properties of water sample
CO4 - Perform quantitative investigation on carbon-based energy source.
CO5 - Analyse the demand oxygen value of water sample
CO6 - Evaluate the physical significance of Chemistry practical towards Petroleum Industry

TEXT / REFERENCE BOOKS

6. Guttag, J. V. (2013). *Introduction to computation and programming using Python*. Prentice Hall.
7. Downey, A. B., Elkner, J., & Meyers, C. (2002). *How to think like a computer scientist: Learning with Python*. Green Tea Press.
8. Sweigart, A. (2015). *Automate the boring stuff with Python: Practical programming for total beginners*. No Starch Press.
9. Brown, M. C. (2018). *Python: The complete reference* (4th ed.). McGraw-Hill Education.
10. Rao, R. N. (2018). *Core Python programming*. Dreamtech Press.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24PE102T					Sedimentary and Petroleum Geology					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	-	-	100

COURSE OBJECTIVES

- Demonstrate the fundamentals of Sedimentology in Reservoir characterization
- Understand the geological process involved in formation of Hydrocarbons
- Imbibe the knowledge of petroliferous basins of India
- Improve analytical skills to interpret the geological history of a Petroliferous basin.

Unit I: **Hours: 10**

Basins and formation of sediments, Grain transportation and its deposition, Hjoulstorm curve, Sediment texture (Size, Shape, sorting); Control of grains size on Porosity and permeability; Carbonate rock and its porosity; Paleocurrent analysis, diagenesis, fundamentals of fluid flow (Reynolds no; Froude no.); Flow Regimes (Upper flow regime; Lower flow regime); Introduction to Unidirectional flows and oscillatory flows.

Unit II: **Hours: 10**

Bedforms formed under unidirectional flow; Bedforms formed under Oscillatory flow; Sediment stability diagram; Interpreting Depositional environment from outcrops and cores. Interpretation of Sedimentary environment especially, Sedimentology (Grain size, sorting), Sand distribution, Sand geometry, Porosity-permeability trends in reservoirs formed in Continental (Alluvial Fan, Fluvial System-Meandering and Braided); Delta and Estuary System; Shallow marine (Shoreface); Shelf and Deep and Ultra deep sedimentary environments. Selected case studies of each environment.

Unit III: **Hours: 09**

Theory of Petroleum formation; Organic matter and its composition; Kerogen and its types, Maturation of Kerogene, Thermal maturity indicators, Migration of Hydrocarbons; Primary migration-Theories and reasons; Secondary migration; Tertiary migration; Migration under hydrostatic and hydrodynamic conditions; effect of isopotential surfaces, identifying migration pathways.

Unit IV: **Hours: 10**

Distinction between reservoir and source rock; Source rock characterization; Seal and Cap rock characterization; Traps (Structural; Stratigraphic and combined traps); Total petroleum System; Evaluation of a petroliferous sedimentary basin; Petroleum Geology of India; Selected Case study on (a) Carbonate Reservoir/field –Ghawar; Mumbai-High (b) Clastic reservoir: Assam, Cambay (c) Evaporitic reservoir: Bikaner-Nagur Basin.

Max <39Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recognize, describe and classify the different types of sediments and sedimentary rocks and petroleum components.
CO2 - Comprehend the various types of reservoir sedimentology and sand geometry for hydrocarbon system.
CO3 - Evaluate the petrophysical properties of the sedimentary rocks, and estimate the reservoir potential and evaluate the seal integrity.
CO4 - Evaluate the petroleum source rocks, and estimate the petroleum potential of the petroleum system(s) in a sedimentary basin.
CO5 - Analyze, source rocks, reservoir rocks, migration paths, and traps in petroliferous basins.
CO6 - Analyze and comprehend different components of the Total petroleum system.

TEXT / REFERENCE BOOKS

1. Slatt, R. M. (2006). *Stratigraphic reservoir characterization for petroleum geologists, geophysicists, and engineers* (2nd ed.). Elsevier.
2. Selley, R. C., & Sonnenberg, S. A. (2014). *Elements of petroleum geology* (3rd ed.). Academic Press.
3. Boggs, S., Jr. (1987). *Principles of sedimentology and stratigraphy*. Merrill Publishing Company.
4. Reineck, H. E., & Singh, I. B. (1980). *Depositional sedimentary environments*. Springer.
5. Tissot, B. P., & Welte, D. H. (1984). *Petroleum formation and occurrence: A new approach to oil and gas exploration*. Springer.
6. North, F. K. (1985). *Petroleum geology* (2nd ed.). Unwin Hyman.
7. Beaumont, E. A., & Foster, N. H. (Eds.). (1999). *Treatise of petroleum geology/Handbook of petroleum geology: Exploring for oil and gas traps*. American Association of Petroleum Geologists.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24PE102P					Sedimentary and Petroleum Geology Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- Demonstrate the need for studying reservoir rocks and maps
- Demonstration the way to prepare various maps
- Improve skills to study subsurface structure
- Support in opting for more advanced study of reservoir modelling and basement mapping

LIST OF EXPERIMENTS

1. Study of Sedimentary rocks in Hand Specimen: Sandstone- Detailed study of Different types of Sandstone, Ex. Arkose, Wacke with respect to texture (grain size, shape, and angularity/roundness), sorting, presence/absence of cement/matrix), porosity, permeability, and their depositional environment.
2. Shale- Detailed study of different types of shales (glauconitic, fossiliferous, and carbonaceous) & their depositional environment
3. Limestone-Detailed study of different types of Limestone and their depositional environment
4. Study of Shape of pebbles
5. Grain size of analysis of sediment (Mean, Median, Standard deviation, Skewness, Kurtisis)
6. Study of thin sections of sedimentary rocks (Shale, Sandstones, limestones) vis-à-vis their porosity and permeability.
7. Study of Isochore maps and construction of sections
8. Study of Isopach maps and construction of sections
9. Construction of panel and fence diagrams

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand different minerals in hand specimen.
CO2 - Analyze silicates in hand specimen.
CO3 - Analyze fossils in hand specimen
CO4 - Understand different types of rocks in hand specimen
CO5 - Study clinographic projection of cube
CO6 - Interprate topographic and geological maps

TEXT / REFERENCE BOOKS

1. Sengupta, S. M. (1996). *Introduction to sedimentology*. Oxford & IBH Publishing.
2. Bjørlykke, K. (2010). *Petroleum geoscience: From sedimentary environments to rock physics*. Springer.
3. Pettijohn, F. J. (1975). *Sedimentary rocks* (3rd ed.). Harper & Row.
4. Boggs, S., Jr. (2006). *Principles of sedimentology and stratigraphy* (4th ed.). Pearson Prentice Hall.
5. Reineck, H. E., & Singh, I. B. (1980). *Depositional sedimentary environments*. Springer.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24HS103T					Indian Knowledge System					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- To connect with the tradition of IKS.
- The students will be able to define Philosophical foundations of IKS.
- To understand Fundamentals of Art and Architecture, Traditional and Historical Town Planning & Ancient Indian Art & Architecture.

Unit I: Introduction to IKS **Hours: 07**

What is IKS? Historical and philosophical foundations , Knowledge Framework and classifications in IKS, Indian scheme of knowledge, The knowledge triangle, Framework for establishing valid knowledge, Deductive or inductive logic framework, Potential fallacies in the reasoning process

Unit II: IKS and Engineering Disciplines **Hours: 07**

Engineering marvels of ancient India, Civil engineering: Urban planning, water management, Materials engineering: Sustainable materials, traditional construction techniques Material science and Metallurgy in IKS, Mechanical engineering: Robotics and automation in ancient India, The role of IKS in addressing climate change challenge, IKS principles for resource management and conservation

Unit III: IKS and Mathematics **Hours: 06**

IKS and Indian Mathematics, Algebra, Geometry, Trignometry, Binary Mathematics, Magic, IKS and Indian Astronomy, Pañcāṅga – The Indian calendar system, Astronomical Instruments -Yantras

Unit IV: IKS and Other Disciplines **Hours: 06**

Indian Administration, Concept of state, Anushashan Parwa of Mahabharat, Kautilya's Arthshastra, Social and Political Philosophies of Ancient India , IKS and Medicine: Ayurveda - A Holistic Approach to Health and Wellbeing

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the fundamental concepts of Indian Knowledge System.
CO2 - Describe various cost concepts of IKS relevant from engineering perspective.
CO3 - To Apply Mathematics in the Vedas and Śulva SūtraSystems employed for representing numbers Spherical trigonometry &Celestial Sphere;
CO4 - To Analyse the science of Astronomy and the different units of time discussed in the texts;
CO5 - To be able to weigh the IKS systems vis s vis the current applications in various spheres of understanding;
CO6 - To design applications and processes by incorporating thebl traditional knowledge to the present day concerns.

TEXT / REFERENCE BOOKS

1. Mahadevan, B., Bhat, V. R., & Nagendra, P. (2022). *Introduction to Indian knowledge system: Concepts and applications*. PHI Learning.
2. Chandrasekharendra Saraswati, S. (2005). *The Vedas*. Bharatiya Vidya Bhavan.
3. Singh, A. K., & Kapoor, K. (Eds.). (2005). *Indian knowledge systems* (Vols. 1–2). D. K. Printworld.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24HS104T					Organizational Behaviour					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

- To enable the development of a perspective to diagnose & efficiently deal with matters of human behaviour in organisations.
- To enrich understanding of the dynamics of interaction and integration between the individual and the organization.
- To explore the organization system in entirety to capture the full complexity of organizational behaviour.
- Critically appraise the potential effects of important developments in the external environment on organizational behaviour

Unit I: Introduction to Organizational Behaviour **Hours: 04**

Definition and scope of organization, behaviour, and organizational behaviour, Historical developments and emerging concerns in OB, Perception in organizational behaviour: elements, principles, and biases, Learning theories, conditioning, application in organizations, Personality theories, Attitudes in the workplace

Unit II: Emotions, Stress Management, Group Dynamics and Interpersonal Relationships **Hours: 03**

Understanding emotions in organizational behaviour, Emotional labor, Stress management techniques in organizational settings, Group formation and dynamics in organizations, Group decision-making techniques and their application, Team building strategies and effective communication, Influence, power dynamics, and organizational politics

Unit III: Leadership and Organizational Culture **Hours: 03**

Leadership theories: overview and application in organizational settings, Leadership styles and their impact on organizational behaviour, Understanding organizational culture and climate

Unit IV: Organizational Change and Structure, Balancing Work and Life **Hours: 03**

Organizational change processes, Resistance management during organizational change, Organizational structure and design, Job satisfaction: determinants and impact on employee behaviour, Stress prevention and management techniques, Work-life balance strategies and their role in organizational behaviou

Max <13Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Critically gage the potential effects of important developments in the external environment on organizational behaviour.
CO2 - Analyze organizational behavioural issues in the context of organizational behaviour theories, models, and concepts.
CO3 - Analyze the behaviour of individuals and groups in organizations in terms of the crucial factors that effect organization behaviour.
CO4 - Analyze organizational behavioural matters in the context of organizational behaviour theories, models, and concepts.
CO5 - Demonstrate how the organizational behaviour can integrate in understanding the motivation (why) behind behaviour of people in the organization.
CO6 - Manage conflict in organizational context and deal with stress.

TEXT / REFERENCE BOOKS

1. Wilson, F. M. (2018). *Organizational behaviour and work* (5th ed.). Oxford University Press.
2. Robbins, S. P., Judge, T. A., & Vohra, N. (2019). *Organizational behavior* (18th ed.). Pearson Education.
3. Aswathappa, K. (2019). *Organizational behavior* (12th ed.). Himalaya Publishing House.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**
PART A: 10 Questions of 2 marks each-No choice **20 Marks**
PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24HS105T					Professional Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	-	-	100

COURSE OBJECTIVES

- To strengthen the communication skills of professionals to make them ready for the modern workplace.
- To fine tune their professional skills and expertise using communication skills.
- To participate in the life-long learning process with confidence and certainty.
- To develop analytical, research, and organizational skills through communication skills for a fulfilling career.

Unit I: Technical Writing **Hours: 06**

E-mails, Report Writing, Proposal writing (Project, Research and Business), Review Writing (Books and Articles)

Unit II: Oratory Skills **Hours: 06**

Art of introducing oneself, Presentation Skills, Public speaking

Unit III: Employability Skills **Hours: 07**

Deciphering JD (Job Descriptions), Cover Letter Drafting, Resume Writing, Group Discussion, Interview Skills

Unit IV: Advanced Reading and Critical Thinking **Hours: 07**

Critical Thinking, Language and Communication, Ethical Considerations, Interdisciplinary Perspective, Communication Skills, Philosophy of Science

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Develop the technical writing skills essential for effective communication in professional settings, including business environments, academic institutions, and research organizations.

CO2 - Master the art of public speaking by delivering clear, engaging, and persuasive presentations to diverse audiences, demonstrating confidence, poise, and effective communication techniques.

CO3 - Demonstrate the ability to decipher job descriptions, extracting key requirements, and qualifications to align their skills and experiences effectively.

CO4 - Effectively prepare for and participate in group discussions, demonstrating active listening, critical thinking, and the ability to articulate ideas coherently and persuasively.

CO5 - Understand the importance of clear, honest communication in engineering practice and ethical decision-making in upholding privacy rights and societal well-being.

CO6 - Explore interdisciplinary approaches to problem-solving

TEXT / REFERENCE BOOKS

1. Kumar, K. B. (2022). *Effective communication skills*. Khanna Book Publishing.
2. Wood, F. T. (2007). *Remedial English grammar*. Macmillan.
3. Sharma, S., & Mishra, B. (2009). *Communication skills for engineers and scientists*. PHI Learning Pvt. Ltd.
4. Kaul, A. (2006). *Business communication*. Prentice Hall of India.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

PART A: 10 Questions of 2 marks each-No choice **20 Marks**

PART B: Each unit carries 20 marks, with internal choice **80 Marks**

24YOG101					Yoga, Health & Hygiene					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To impart the students with the basic concepts of physical education, sports, and yoga for health and wellness.
- To familiarize the students with health-related exercises, sports, and yoga for overall growth and development.
- To create a foundation for professionals in physical education, sports, and yoga.
- To impart the basic knowledge and skills to teach physical education, sports, and yoga activities.

ACTIVITY I: KINESIOLOGY AND CARDIO FITNESS TEST

Introduction to Kinesiology and the Physiological Basis of Conditioning, Sports Psychology, and the Cooer Cardio Fitness Test 12-Minutes Run/Walk: How to Start Walking/Runing: Get expert tips, tools, and training.

ACTIVITY II: YOGA

Introduction to Yogasana and Yoga Therapy: A Rehabilitation Tool and the Effect of Yoga on Exercise Endurance as Assessed by Cardiorespiratory Efficiency Tests: Studt on yogic practices that promote and improve respiratory and cardiovascular function
and enhance physical fitness.

ACTIVITY III:GAMES AND SPORTS LEAGUE

Practice sessions for outdoor and indoor games, event-wise practice, and team games organized on the sport, game-wise practice as per the student's interest: Football, Chess, Cricket, Tennis, Basketball, Volleyball, Athletics (Relay), Pickleball.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 -Discover the balance of health and happiness through the basic principles and practices of physical education, sports, and yoga

CO2 -Understand that the students will be able to be instructed on physical activities, sports, yoga practices, theories, and rules of various games for healthy living.

CO3-Analyze emerging trends and issues in world sports and develop leadership qualities among students to conduct, organize, and officiate physical education, sports, and yoga events at schools, colleges, and the community.

CO4 - Practice on the field and in the indoor yoga hall.

CO5 - Develop a spirit of teamwork and fair play

CO6 - Demonstrate understanding by participating in games and sports leagues.

TEXT / REFERENCE BOOKS

1. Agola, R. (n.d.). *Athletic track and court marking handbook of games and sport.*
2. Satyananda Saraswati, S. (n.d.). *Asana, pranayama, mudra, bandha* (or *Asana, pranayama, and kriyas*, if that is the exact title of your edition). Bihar School of Yoga, Munger.
3. Pathak, P. V. (n.d.). *Sports games and rule, regulation*.
4. Gharote, M. L. (n.d.). *Yogic prakriyanche margdarshan*. The Lonavala Yoga Research Institute.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**

24NSS101					National Service Scheme (NSS)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To develop a sense of civic and social responcibility.
- To identify the needs and problems of the community and involve them in problem-solving.
- To engage in creative and constructive social action.
- To develop social character and leadership through NSS.

ACTIVITY I: ENVIRONMENT AND SUSTAINABILITY -WATER - WAST MANAGEMENT
Volunteering work for Environment & Sustainability (water and waste management) and Tree Plantation.

ACTIVITY II: NSS 7 DAYS SPECIAL CAMP
Volunteering for tree planting, agriculture compost, tree guard,Gujarat Skill Development Mission, and social activities in the village as per the government NSS manual.

ACTIVITY III: FIT INDIA MISSION
Volunteering for Cardio Fitness, Yoga, Running, Mission Olympics, Self-Defense, and Agneepath Mission.

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to
CO1 -Identify the needs and problems of the community.
CO2 -Understand the importance of his / her responsibilities towards society.
CO3- Analyze the environmental and societal problems/issues.
CO4 –Evaluate the existing system and propose practical solutions for the sustainable development.
CO5 -Develop a government or self-driven projects effectively in the field.
CO6 -Understand the government or self-driven projects effectively in the society.

TEXT / REFERENCE BOOKS

1. Pandit Deendayal Energy University, National Service Scheme Unit. (2024). *NSS course manual*. Pandit Deendayal Energy University. https://www.pdpu.ac.in/nssreport.html
2. National Service Scheme, Government of Gujarat. (n.d.). *Gujarat State NSS Cell*. Ministry of Youth Affairs and Sports, Government of India. https://nss.gov.in/gujarat-1
3. National Service Scheme, Ministry of Youth Affairs and Sports, Government of India. (n.d.). *National Service Scheme: Activities, reports, and manuals*. https://nss.gov.in/

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**
End semester examination and Viva-voce **50 Marks**

24NCC101					National Cadet Corps (NCC)					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	-	-	-	50	50	100

COURSE OBJECTIVES

- To develop discipline, character, brotherhood, the spirit of adventure and ideals of selfless service amongst young students.
- To develop youth leadership qualities in the students.
- To induce social consciousness among students through various NCC camps

ACTIVITY I: INTRODUCTION TO NCC

Introduction to NCC, aims and objectives, structure and organization of NCC, cardinals of NCC, NCC Flag, oath of NCC, NCC Song, incentives of NCC

ACTIVITY II: NATIONAL INTEGRATION

Importance of national integration and awareness, necessity, national interests, objectives, threats and opportunities, unity in diversity

ACTIVITY III: NCC CAMPS, SOCIAL SERVICE, AWARENESS AND COMMUNITY DEVELOPMENT ACTIVITIES

Social awareness & community development, health & hygiene, environment awareness and conservation, cadets will participate in various activities e.g., blood donation camp, swachhata abhiyan, constitution day, etc., participation into NCC camps like ATC, CATC, NIC, COC, TSC, RDC, leadership camps, etc.

ACTIVITY IV: DRILL, WEAPON TRAINING AND ADVENTURE ACTIVITIES

Types of drill, foot drill, general and words of command, saluting, weapon training, map reading, field craft & battle craft, Introduction to infantry weapons & equipment, obstacle and weapon training (during camps), adventure training, participation into Republic and Independence day ceremonial parades at university

Max <26Hrs>

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 -Know about the history of NCC, its organization, and incentives of NCC for their career prospects.
CO2 -Understand the importance of Nation building and individual contribution to the same.
CO3- Maintain discipline and team spirit
CO4 –Build the character and leadership qualities
CO5 -Understand that drill as the foundation for discipline and to command a group for common goal.
CO6 -Develop the sense of self-less social service for better social & community life.

TEXT / REFERENCE BOOKS

1. Directorate General, National Cadet Corps. (2017). *Cadet's handbook: SD/SW common subjects (All wings).* Ministry of Defence, Government of India.

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs.**

Evaluation based on the class performance and Laboratory book **50 Marks**

End semester examination and Viva-voce **50 Marks**